

EDGE-DELETION COUNTEREXAMPLES TO THE ALIABADI–KROP SHARP UNIFORM-GAP FORMULA

ABSTRACT. Aliabadi and Krop asked for the sharp normalized perfect-matching count below the Alon–Friedland bound among non-extremal graphs of maximum degree at most Δ . For every $\Delta \geq 4$, we show that deleting one edge from $K_{\Delta, \Delta}$ gives a ratio larger than both candidates in their proposed formula. The same family yields an upper bound of order $(\log \Delta)/\Delta^2$ on any uniform gap constant, improving their order- $1/\Delta$ obstruction.

1. THE COUNTEREXAMPLES

For a finite simple graph G of even order and without isolated vertices, set

$$U(G) = \prod_{v \in V(G)} (d_G(v)!)^{1/(2d_G(v))}.$$

The Alon–Friedland bound states that $\text{perfmat}(G) \leq U(G)$, with equality exactly for disjoint unions of balanced complete bipartite graphs [1].

For $\Delta \geq 3$, Aliabadi and Krop define

$$R_\Delta = \sup \left\{ \frac{\text{perfmat}(G)}{U(G)} : \begin{array}{l} G \text{ has even order, no isolated vertices, and } \Delta(G) \leq \Delta, \\ G \text{ is not a disjoint union of balanced complete bipartite graphs} \end{array} \right\}$$

and ask in [2, Problem 4.9] whether

$$(1) \quad R_\Delta = \max \left\{ \left(\frac{3}{4} \right)^{1/3}, \rho_{\Delta-1} \right\}, \quad \rho_r = \frac{(r!)^{1/(r(r+1))}}{(r+1)^{1/(r+1)}}.$$

For $r \geq 2$, let $H_r = K_{r,r} - e$, where e is one edge, and write

$$\sigma_r = \frac{\text{perfmat}(H_r)}{U(H_r)}.$$

Aliabadi and Krop observed H_3 as the runner-up in their enumeration of connected graphs on at most seven vertices [2, Remark 4.10]. The full deletion family refutes (1) from $r = 4$ onward.

Theorem 1. *For every integer $r \geq 4$, the graph H_r is admissible in the definition of R_r and*

$$\sigma_r = \frac{r-1}{r^{(r-1)/r}((r-1)!)^{1/(r(r-1))}} > \max \left\{ \left(\frac{3}{4} \right)^{1/3}, \rho_{r-1} \right\}.$$

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Consequently, the formula (1) fails for every $\Delta \geq 4$.

Proof. The graph H_r is connected, has even order, has no isolated vertices, and has maximum degree r . Since one cross-edge is missing, it is not a balanced complete bipartite graph. Of the $r!$ perfect matchings of $K_{r,r}$, exactly $(r-1)!$ use the deleted edge. Hence

$$\text{permat}(H_r) = (r-1)(r-1)!.$$

There are $2r-2$ vertices of degree r and two of degree $r-1$, so

$$U(H_r) = (r!)^{(r-1)/r} ((r-1)!)^{1/(r-1)},$$

which gives the stated formula for σ_r .

By strict arithmetic–geometric mean,

$$(2) \quad ((r-1)!)^{1/(r-1)} < \frac{r}{2}.$$

Therefore

$$\frac{\sigma_r}{\rho_{r-1}} = \frac{r-1}{r^{(r-2)/r} ((r-1)!)^{2/(r(r-1))}} > \left(1 - \frac{1}{r}\right) 2^{2/r} > 1.$$

For the last inequality, the function $x \mapsto x \log(1 - 1/x)$ is increasing on $(1, \infty)$, since its derivative is

$$\frac{1}{x-1} - \log\left(1 + \frac{1}{x-1}\right) > 0.$$

Thus, for $r \geq 3$,

$$\left[\left(1 - \frac{1}{r}\right) 2^{2/r}\right]^r = 4 \left(1 - \frac{1}{r}\right)^r \geq 4 \left(\frac{2}{3}\right)^3 > 1.$$

It remains to compare σ_r with $(3/4)^{1/3}$. From (2),

$$\sigma_r > q(r) := \left(1 - \frac{1}{r}\right) 2^{1/r}.$$

The function q is increasing on $(1, \infty)$ because

$$\frac{d}{dx} \log q(x) = \frac{x/(x-1) - \log 2}{x^2} > 0.$$

Moreover,

$$\left(\frac{q(5)}{(3/4)^{1/3}}\right)^{15} = \frac{2^{43}}{3^5 5^{15}} > 1,$$

where the final inequality follows from $3^5 < 2^8$ and $5^3 < 2^7$. Hence $\sigma_r > (3/4)^{1/3}$ for every $r \geq 5$. For $r = 4$, direct simplification gives

$$\left(\frac{\sigma_4}{(3/4)^{1/3}}\right)^{12} = \frac{3^7}{2^{11}} > 1.$$

This proves the two strict comparisons. Taking $r = \Delta$ completes the proof. $\square$

2. A SHARPER ASYMPTOTIC OBSTRUCTION

Aliabadi and Krop obtained

$$c(\Delta) \leq \frac{1}{\Delta} + O\left(\frac{\log \Delta}{\Delta^2}\right)$$

from an added-edge construction [2, Corollary 4.8]. The deletion family gives the following stronger obstruction.

Corollary 2. *As $\Delta \rightarrow \infty$,*

$$1 - R_\Delta \leq 1 - \sigma_\Delta = \frac{\log(2\pi\Delta) - 1}{2\Delta^2} + O\left(\frac{\log \Delta}{\Delta^3}\right).$$

In particular, any constant $c(\Delta)$ valid in [2, Problem 4.4] satisfies the same upper bound.

Proof. From the formula in Theorem 1,

$$\log \sigma_r = \log\left(1 - \frac{1}{r}\right) + \frac{\log r}{r} - \frac{\log \Gamma(r)}{r(r-1)}.$$

Stirling's formula gives

$$\log \sigma_r = -\frac{\log(2\pi r) - 1}{2r^2} + O\left(\frac{\log r}{r^3}\right).$$

The square of the main term is $o((\log r)/r^3)$, so exponentiation yields

$$1 - \sigma_r = \frac{\log(2\pi r) - 1}{2r^2} + O\left(\frac{\log r}{r^3}\right).$$

Since $R_\Delta \geq \sigma_\Delta$, the result follows. $\square$

Remark 3. The theorem does not determine R_Δ , settle the case $\Delta = 3$, or resolve whether $R_\Delta < 1$ for each fixed Δ .

REFERENCES

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